

Agent Decision Loop

Objective

The aim of this task is to design a simple AI agent that can make decisions step by step until a task is completed. The agent checks the current situation, decides what to do next, and stops when the required condition is achieved.

Task: Hotel Room Booking

The AI agent is designed to help a guest complete a hotel room booking.

Decision Loop

The agent follows this process:

Guest Request -> Check Information -> Decide Next Action -> Perform Action -> Check Result -> Repeat or Stop

Step-by-Step Process

1. Receive the guest's request
2. The guest asks to book a hotel room.
3. Check the required information

The agent checks whether it has the guest's name, dates, number of guests, and room type.

4. Make a decision
 - a. If information is missing → ask the guest for it.
 - b. If all information is available → check room availability.
5. Check availability
 - a. If the room is available → proceed with the booking.
 - b. If it is unavailable → suggest another available room or date.
6. Confirm the booking

Once a suitable room is available, the agent confirms the booking with the guest.

Stopping Condition

The agent stops when the booking is successfully confirmed.

It should also stop if the guest cancels the request or if no suitable room is available after the defined number of attempts.

Simple Example

Guest: I want to book a Deluxe room.

Agent: What dates would you like to stay?

Guest: 15 to 17 September.

Agent: How many guests?

Guest: Two.

Agent: Checks availability → Room available → Confirms booking.

STOP: Booking confirmed.

Key Learning

Through this task, I learned how an AI agent can make decisions step by step instead of giving just one response. I also understood why a clear stopping condition is important. It prevents the agent from continuing unnecessarily and ensures that the task ends once the required result is achieved.